PAS Table 3, recomputed after the estimator and pseudo-ground-truth fixes

$K = 200$ replicates, default split seed (42; replicate k uses seed $42 + k$), `train_test_split` $= 0.2$.

Table 1: Post-fix results aggregated over $K = 200$ replicates on the three real-world datasets: Amazon review ratings with `BERT-base` and `BERT-tuned` predictors, and spiral galaxy fractions with `ResNet50` predictor. **Second moments estimated per problem (`share_var=False`).** Metrics are reported with $\pm$ 1 standard error.

	Amazon (base f)		**Amazon (tuned f)**		**Galaxy**	
Estimator	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$
Classical	15.607 ± 0.121	baseline	15.607 ± 0.121	baseline	1.361 ± 0.016	baseline
Prediction Avg	41.941 ± 0.057	23.6 ± 0.2	4.697 ± 0.023	69.3 ± 0.2	7.252 ± 0.011	13.0 ± 0.2
PPI	9.973 ± 0.072	56.8 ± 0.2	6.262 ± 0.047	64.4 ± 0.2	1.024 ± 0.013	54.0 ± 0.3
PT	6.663 ± 0.056	67.6 ± 0.2	4.702 ± 0.039	73.1 ± 0.2	0.645 ± 0.009	65.1 ± 0.3
Shrink Classical	10.202 ± 0.072	54.3 ± 0.3	2.786 ± 0.021	77.3 ± 0.3	0.987 ± 0.010	50.2 ± 0.4
Shrink Avg	6.526 ± 0.058	67.8 ± 0.2	4.670 ± 0.040	73.1 ± 0.2	0.645 ± 0.009	65.9 ± 0.4
`PAS` (ours)	5.588 ± 0.045	**68.7 ± 0.2**	2.758 ± 0.024	77.7 ± 0.2	0.581 ± 0.008	**66.3 ± 0.4**
`UniPT` (ours)	6.214 ± 0.051	67.2 ± 0.2	3.678 ± 0.031	73.3 ± 0.2	0.622 ± 0.008	65.3 ± 0.4
`UniPAS` (ours)	**5.475 ± 0.044**	67.8 ± 0.2	**2.071 ± 0.016**	**78.3 ± 0.2**	**0.546 ± 0.006**	65.9 ± 0.3

Table 2: Post-fix results aggregated over $K = 200$ replicates on the three real-world datasets. **Second moments pooled across problems (`share_var=True`).** Only PT, Shrink Classical, Shrink Avg and PAS differ from Table 1. Metrics are reported with $\pm$ 1 standard error.

	Amazon (base f)		**Amazon (tuned f)**		**Galaxy**	
Estimator	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$
Classical	15.607 ± 0.121	baseline	15.607 ± 0.121	baseline	1.361 ± 0.016	baseline
Prediction Avg	41.941 ± 0.057	23.6 ± 0.2	4.697 ± 0.023	69.3 ± 0.2	7.252 ± 0.011	13.0 ± 0.2
PPI	9.973 ± 0.072	56.8 ± 0.2	6.262 ± 0.047	64.4 ± 0.2	1.024 ± 0.013	54.0 ± 0.3
PT	6.222 ± 0.051	66.6 ± 0.2	3.684 ± 0.030	72.9 ± 0.2	0.624 ± 0.008	64.7 ± 0.4
Shrink Classical	10.427 ± 0.079	53.6 ± 0.3	2.679 ± 0.035	75.4 ± 0.3	1.008 ± 0.012	46.7 ± 0.4
Shrink Avg	6.079 ± 0.052	66.5 ± 0.2	3.727 ± 0.031	72.4 ± 0.2	0.645 ± 0.009	64.9 ± 0.4
`PAS` (ours)	5.515 ± 0.044	67.3 ± 0.2	2.126 ± 0.016	78.0 ± 0.2	0.581 ± 0.007	63.9 ± 0.3
`UniPT` (ours)	6.214 ± 0.051	67.2 ± 0.2	3.678 ± 0.031	73.3 ± 0.2	0.622 ± 0.008	65.3 ± 0.4
`UniPAS` (ours)	**5.475 ± 0.044**	**67.8 ± 0.2**	**2.071 ± 0.016**	**78.3 ± 0.2**	**0.546 ± 0.006**	**65.9 ± 0.3**

The MSE scale changed; these tables are not comparable row-for-row with the published Table 3. The pseudo-ground-truth now averages *all* labels for a problem (Appendix E.4, $\dot{\theta}_j := T_j^{-1} \sum_i \dot{Y}_{ij}$) instead of the unlabelled split only. Because $\bar{Y}_{\text{lab}} - \bar{Y}_{\text{all}} = (N/T)(\bar{Y}_{\text{lab}} - \bar{Y}_{\text{unlab}})$ pointwise, every squared error is rescaled by $(N/T)^2 = 0.641$ at an 80% unlabelled split; the measured Classical ratio is 0.6417. Relative orderings carry over, absolute levels do not.

Which rows respond to `share_var`. Only PT, Shrink Classical, Shrink Avg and PAS take the flag. Classical, Prediction Avg and PPI have no second-moment estimate to share, and UniPT/UniPAS deliberately omit it (the paper notes sharing variance is not meaningful once a single global λ is used), so those five rows are identical across Tables 1 and 2. **Bolding** marks the best value in each column.

Table 3: *Reference only:* the published Table 3, computed before the fixes. The MSE column scales are not comparable with Tables 1–2 because the pseudo-ground-truth changed; see the note on the preceding page.

	Amazon (base f)		**Amazon (tuned f)**		**Galaxy**	
Estimator	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$	**MSE** ($\times 10^{-3}$)	**% Improved** $\uparrow$
Classical	24.305 ± 0.189	baseline	24.305 ± 0.189	baseline	2.073 ± 0.028	baseline
Prediction Avg	41.332 ± 0.050	30.7 ± 0.2	3.945 ± 0.011	75.4 ± 0.2	7.195 ± 0.008	17.0 ± 0.2
PPI	11.063 ± 0.085	62.4 ± 0.2	7.565 ± 0.066	70.4 ± 0.2	1.149 ± 0.017	59.4 ± 0.3
PT	10.633 ± 0.089	70.3 ± 0.2	6.289 ± 0.050	76.0 ± 0.2	1.026 ± 0.015	67.7 ± 0.3
Shrink Classical	15.995 ± 0.121	56.4 ± 0.3	3.828 ± 0.039	78.9 ± 0.2	1.522 ± 0.016	48.8 ± 0.4
Shrink Avg	9.276 ± 0.078	70.4 ± 0.2	6.280 ± 0.058	77.1 ± 0.2	0.976 ± 0.014	**68.9 ± 0.3**
PAS (ours)	**8.517 ± 0.071**	**71.4 ± 0.2**	**3.287 ± 0.024**	**80.8 ± 0.2**	**0.893 ± 0.011**	67.3 ± 0.4
UniPT (ours)	10.272 ± 0.084	70.0 ± 0.2	6.489 ± 0.053	76.2 ± 0.2	1.017 ± 0.015	67.7 ± 0.3
UniPAS (ours)	8.879 ± 0.073	69.5 ± 0.2	3.356 ± 0.031	77.6 ± 0.2	0.909 ± 0.011	66.7 ± 0.3